

Setup and Model Economy

We will study the welfare state through the lens of a dynamic labor supply model. Baseline code is available via Digital Exam. Workers live for T periods and $t \in \{1, 2, \dots, T\}$ thus denotes their age. In each period, individuals choose how much to work, $l_t \in [0, 1]$, generating labor earnings. Wages, w_t , are endogenous through human capital accumulation. We denote $k_t \in \mathbb{R}_+$ as the level of human capital. Individuals receive unemployment benefits of π_t if not working. Individuals differ by their birth cohort, $b \in \{0, 1, 2, 3, 4\}$ and $y_t = t + b \in \{1, 2, \dots, T + 4\}$ tracks the year in the model. Workers can experience aggregate shocks through a process, $\lambda(y_t)$, that scales all individual wages equally in a given year.

The recursive formulation (Bellman equation) for the model is, for $1 \leq t < T$,

$$V_t(b, k_t) = \max_{l_t \in [0, 1]} U(c_t, l_t) + \beta \mathbb{E}_t[V_{t+1}(b, k_{t+1})] \quad (1)$$

s.t.

$$k_{t+1} = [(1 - \delta)k_t + l_t]\varepsilon_{t+1} \quad (2)$$

$$c_t = (1 - \tau)w_t l_t + \pi_t \cdot \mathbf{1}(l_t = 0) \quad (3)$$

$$w_t = \exp(\alpha_b + \gamma k_t) \cdot \lambda(y_t) \quad (4)$$

$$y_t = t + b \quad (5)$$

$$\lambda(y_t) = 1 + \lambda_0 y_t \quad (6)$$

$$\pi_t = \pi_0 \quad (7)$$

$$\ln \varepsilon_t \sim iid \mathcal{N}(-0.5\sigma^2, \sigma^2) \quad (8)$$

where preferences are given by

$$U(c_t, l_t) = \frac{c_t^{1-\rho}}{1-\rho} - \phi \frac{l_t^{1+\eta}}{1+\eta}. \quad (9)$$

There is no bequest motive such that for $t = T$,

$$V_T(b, k_T) = \max_{l_T \in [0, 1]} U(c_T, l_T) \quad (10)$$

s.t.

$$c_T = (1 - \tau)w_T l_T + \pi_T \cdot \mathbf{1}(l_T = 0). \quad (11)$$

The baseline parameters are $T = 20$, $\beta = 0.97$, $\delta = 0.1$, $\rho = 1.5$, $\gamma = 0.1$, $\phi = 3.0$, $\eta = 2.0$, $\sigma = 0.1$ and $\lambda_0 = -0.001$. The cohort-specific wage parameters are $(\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4) = (0.7, 0.8, 0.9, 1.1, 1.2)$. Let $\theta = (\pi_0, \tau)$ denote the baseline policy parameters in the model and let $\pi_0 = 0.5$ and $\tau = 0.3$.

When simulating agents from the model, we draw initial levels of human capital from a truncated Normal distribution, such that $k_{i,0} = \min\{\max\{\tilde{k}_{i,0}, 0\}, 3\}$ where $\tilde{k}_{i,0} \sim \mathcal{N}(0.5, 1)$. We

assume a uniform distribution of birth cohorts, such that $\Pr(b_i = j) = 1/5$ for all $j \in \{0, 1, 2, 3, 4\}$. We draw individual-specific human capital shocks, $\varepsilon_{i,t}$, from the mean-one, variance σ^2 log-normal distribution. We simulate $N = 50,000$ individuals for $T = 20$ periods. Questions can be answered independently but the idea is that you answer question j building on the setup in question $j - 1$. Try to avoid copy-pasting code but rather generate one code base that can answer all questions.

Questions

1. Explain why we do not need to keep track of the year, y_t , as a separate state in the model.
2. Solve and simulate two versions of the model, one with no labor income taxes, $\tau = 0$, and one with $\tau = 0.3$.
 - (a) Plot the age profiles of average labor supply, l_t , for both models jointly.
 - (b) Plot the age profiles of the employment rate (share with $l_t > 1.0e^{-6}$), for both models jointly.
 - (c) Plot the age profiles of average consumption, for both models jointly.
3. Let $G(\theta) = \sum_i^N \sum_i^T G_{i,t}$ denote the government budget balance for a given set of policy parameters, where $G_{i,t} = \tau w_{i,t} l_{i,t} - \pi_t \cdot \mathbf{1}(l_{i,t} = 0)$ is government tax revenue net of non-employment subsidies. You can store $G_{i,t}$ in “sim” for later use below.
 - (a) What is $G(\theta)$ for the baseline policy parameters, θ ? We refer to this value as G_3 throughout.
 - (b) Plot $G(\theta)$ for $\tau \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$.
 - (c) Discuss the results of this analysis.

Use $\tau = 0.3$ to answer the following questions unless otherwise instructed.

4. Policy makers now consider introducing an old-age retirement subsidy, starting from year 8. Concretely, they want to introduce an *additional* unemployment subsidy of $\pi_1 = 0.6$ for non-working individuals aged 10 or older, starting from year 8 and onward. We will now analyze this reform. Note that the policy parameters now include this parameter such that $\theta = (\pi_0, \pi_1, \tau)$.
 - (a) Formulate this new model, assuming that workers know in advance, that this reform will be implemented, even in years before 8. Be explicit about which equations you modify.

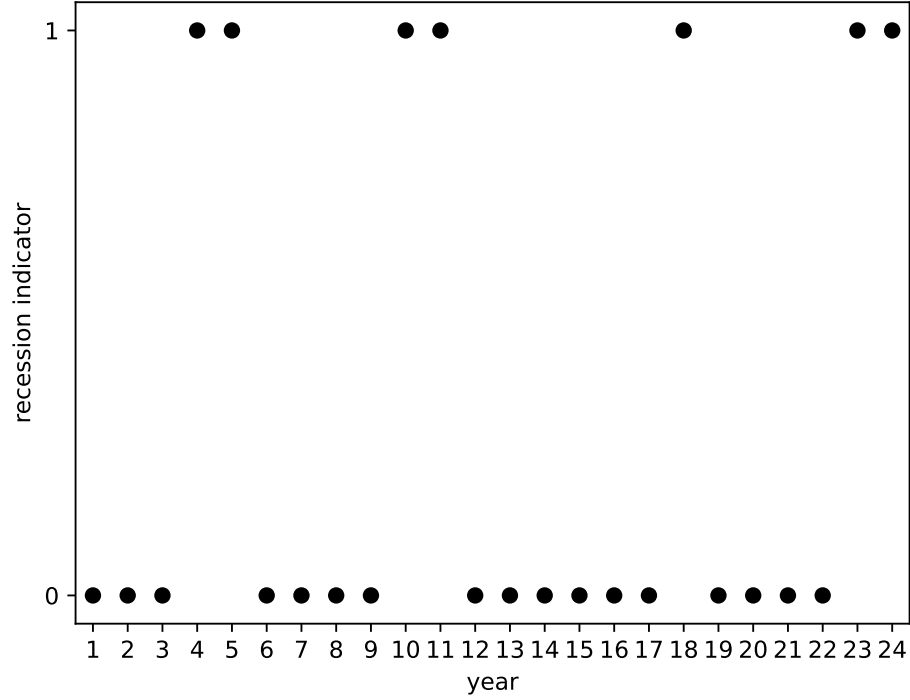
- (b) Solve and simulate this new version of the model. Plot age profiles of average labor supply and the share working from the baseline model in question 3 together with the same output simulated from the new model and discuss potential differences in behavior.
 - (c) Plot the average government budget balance *across years* (not age) for both models.
5. The government is now considering also increasing the tax rate from year 8 onward to finance the old-age retirement reform. Denote this additional tax rate τ_1 and include it in the set of policy parameters, such that $\theta = (\pi_0, \pi_1, \tau, \tau_1)$.
- (a) Introduce this additional tax, τ_1 , in the model. Be precise about which equations you modify.
 - (b) Determine the level of τ_1 that minimizes the effect of the reform on the government budget balance. That is, find τ_1^* that minimizes the difference $G_3 - G(\theta)$, where we denote $\theta_5^* = (\pi_0 = 0.5, \pi_1 = 0.6, \tau = 0.3, \tau_1^*)$ throughout. Search in the interval $[0.015, 0.025]$.
 - (c) Define a measure of worker welfare of your own choice. Be explicit about how you define it mathematically and how you implement it in Python.
 - (d) For each birth-cohort, b , calculate the average welfare measure and the share who are better off under the new partly tax-financed old age reform (that is, with policy parameters set to θ_5^*). Discuss the results.
6. In questions 4-5 above we assumed that workers perfectly foresee the reform change several years before it was proposed to be implemented.
- (a) Discuss how we can use the two versions of the models (the baseline model from question 3 and the alternative in question 4) to simulate a version of the reform in which the reform is unanticipated by the workers, who see the old-age pension reform as a surprise in year 8.
 - (b) Implement the unanticipated reform. Compare average simulated labor supply and human capital accumulation *over time (years)* from *i)* the anticipated reform in question 4 and *ii)* the unanticipated reform. Show differences relative to the baseline model from question 3 and set $\tau_1 = 0$ in all models.
 - (c) Discuss the labor market behavior across the two models.
7. We now want to extend the model to account for the effect of aggregate recessions in the economy. Concretely, we now let $\lambda(y_t, r_t) = \lambda_0 + \lambda_1 r_t$ where $r_t = 1$ denotes a recession period and $r_t = 0$ denotes a non-recession period. The transition probabilities are given in Table 1.

Table 1: Recession Transition Probabilities, $\Pr(r_{t+1}|r_t)$.

	$r_t = 0$	$r_t = 1$
$r_{t+1} = 0$	0.7	0.6
$r_{t+1} = 1$	0.3	0.4

- (a) Write up the new recursive formulation of the model.
- (b) Solve and simulate the model with $\lambda_1 = -0.5$. We will assume that all workers experience the same economy in a given year. That is, in the simulation, we will assume that all workers in year y experience the same r_y , given by
 $[0\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 1\ 1]$, also plotted in Figure 1.

Figure 1: Recession Indicator over Time/Years.



- (c) Plot average labor supply, consumption and recession status over the life cycle, i.e. as a function of age, t .
- (d) Plot average labor supply, consumption and recession status over time, i.e. as a function of year, y .
- (e) Discuss the potential differences.